



NORTH SYDNEY BOYS HIGH SCHOOL

2010 Year 11 Half Yearly

Mathematics Extension 1

General Instructions

- Working time – 100 minutes
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Mr Ireland
- ☐ Mr Fletcher
- ☐ Mr Lam
- ☐ Mr Rezcallah
- ☐ Mr Trenwith
- ☐ Mr Weiss

Student Name: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	5	6	7	8	Total %
Mark	$\frac{\quad}{12}$	$\frac{\quad}{12}$	$\frac{\quad}{14}$	$\frac{\quad}{12}$	$\frac{\quad}{10}$	$\frac{\quad}{19}$	$\frac{\quad}{6}$	$\frac{\quad}{15}$	$\frac{\quad}{100}$

Question 1 (12 marks)**Marks**

- a) Evaluate $\sqrt[3]{\frac{8.5+15.2}{\pi-2.15}}$ correct to 3 significant figures. 2
- b) Express $0.35\dot{2}$ as a fraction in simplest rational form. Show working. 2
- c) Factorise completely:
- (i) $m^2 - mn + 6m - 6n$ 2
- (ii) $x^5 - 32$ 2
- d) If $x = 3 - \sqrt{2}$, simplify fully $(x - 2)^2$ 2
- e) Solve: $|2x + 1| = 5$ 2

Question 2 (12 marks) Start a new page.

- a) Express $\frac{23\pi}{15}$ in degrees. 1
- b) Evaluate $\operatorname{cosec} 0.8$ correct to two decimal places. 1
- c) Find, in simplest surd form, the value of $\frac{1 + \cos 120^\circ}{\tan 240^\circ}$ 3
- d) If $\cos x = -\frac{2}{3}$, find the exact value(s) of $\tan x$. 2
- e) In triangle ABC, AC = 8 cm, BC = 10 cm and $\angle ABC = 50^\circ$.
-
- Find the possible values of $\angle BAC$ to the nearest degree 3
- f) A triangular airfield is bordered by three straight lines of length 4 km, 5 km and 6 km. If the largest angle in the triangle is 83° , find the area of the airfield. 2

Question 3 (14 marks) Start a new page.

- a) Sketch the following relations on separate number planes, showing all important features (asymptotes, x and y intercepts):

(i) $y = \sqrt{25 - x^2}$ 2

(ii) $y = |3 + 2x|$ 2

(iii) $y = \frac{-3}{x+2}$ 2

(iv) $y = 2^{-x} - 1$ 2

- b) (i) Sketch a graph of $f(x) = 4x - x^2$ showing all main features. 2

Use this graph to answer the following questions:

(ii) Find the largest possible domain and range of $y = 4x - x^2$ 2

(iii) Solve $4x - x^2 \geq 0$ 1

(iv) What is the natural domain of the function $H(x) = \frac{3}{\sqrt{4x - x^2}}$ 1

Question 4 (12 marks) Start a new page.

a) Solve the following equations:

$$(i) \quad \frac{x+3}{2} - \frac{3x-1}{7} = 2 \quad 2$$

$$(ii) \quad 5x^2 - 9x = 2 \quad 2$$

$$(iii) \quad |3x + 1| = 4x - 2 \quad 3$$

b) Simplify fully the following:

$$(i) \quad \frac{a^2-9}{a^4-27a} \div \frac{a+3}{a^2+3a+9} \quad 2$$

$$(ii) \quad \frac{x+2}{x-1} - \frac{2x-1}{x+1} - \frac{6x}{x^2-1} \quad 3$$

Question 5 (10 marks) Start a new page.

a) Solve the following inequations:

$$i) \quad |5 - 2x| < 7 \quad 2$$

$$ii) \quad \frac{4x-1}{x-1} \geq 3 \quad 3$$

$$b) \quad \text{Expand and simplify } (2x - 1)^4 \quad 2$$

$$c) \quad \text{Find rational numbers } a \text{ and } b \text{ such that } a + b\sqrt{2} = \frac{\sqrt{2} + 1}{\sqrt{2} - 1} \quad 3$$

Question 6 (19 marks) Start a new page.

a) Solve the following equations for $0 \leq x \leq 360^\circ$:

i) $2 \sin 2x = 1$ 2

ii) $\tan x = \cot x$ 3

b) Prove that $(\tan x + \cot x)^2 = \sec^2 x + \operatorname{cosec}^2 x$. 3

c) Simplify $\tan(90^\circ - \theta) \sin(360^\circ - \theta)$ 2

d) i) State the amplitude, period and the range of $y = 2 \sin 4x$. 3

ii) Sketch the graph of $y = 2 \sin 4x, 0 \leq x \leq 180^\circ$. 2

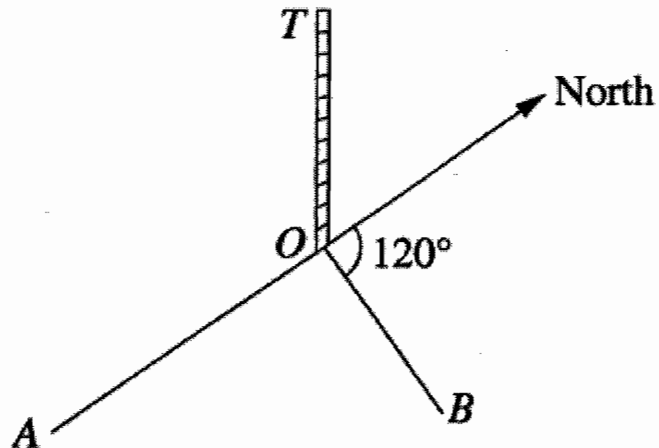
e) A is 3 km North and 4.3 km east of P. The bearing of B from A is 145° .

Find the distance PB if the distance AB = 14 km. 4

Question 7 (6 marks) Start a new page.

From a point A due south of a tower, the angle of elevation of the top of the tower T , is 28° . From another point B , on a bearing of 120° from the tower, the angle of elevation of T is 33° .

The distance AB is 200 metres.



- i) Copy or trace the diagram into your writing booklet, adding the given information to your diagram.

1

- ii) If $OT = h$, express the distances OA and OB in terms of h .

2

- iii) Hence find the height of the tower to the nearest metre.

3

Question 8 (15 marks) Start a new page.

a)

$$f(x) = \begin{cases} \frac{1}{x}, & x \leq 1 \quad x \neq 0 \\ 2x^3 - 1, & x > 1 \end{cases}$$

- (i) Find $f(2)$ 1
- (ii) Sketch the function $f(x)$. 2
- (iii) State the range of $f(x)$. 1

b) If $f(n) = n(n+1)(n+2)$ express $f(n) + f(n+1)$ in factored form. 2

c) If $y = f(x) = \frac{x^2+1}{x^2-1} = 1 + \frac{2}{x^2-1}$

- i) State the equations of the asymptotes. 2
- ii) Show that $y = f(x)$ is an even function. 2
- iii) Sketch the graph showing intercepts and asymptotes. 3
- iv) Hence, or otherwise, solve $\frac{x^2+1}{x^2-1} < 1$ 1

d) Find $g(x)$ if the function $f(x)$ below is an odd function.

$$f(x) = \begin{cases} x^4, & \text{for } x > 0 \\ g(x), & \text{for } x < 0 \end{cases} \quad 1$$

End of Exam

2010 Yr 11 Ext 1 Maths Half Yearly

Question 1

a) $2.880525 = 2.88.$

✓
✓ for sig. figures 2.8

b) let $x = 0.3522$

$10x = 3.5222.$

$\therefore 9x = 3.17$

$x = \frac{3.17}{9} = \frac{317}{900}.$

✓ for $\frac{317}{900}$

c) (i) $m^2 - mn + 6m - 6n$
 $= m(m-n) + 6(m-n)$
 $= (m-n)(m+6).$

✓

✓

(ii) $x^5 - 32 = x^5 - 2^5$
 $= (x-2)(x^4 + 2x^3 + 4x^2 + 8x + 16)$

✓ for $(x-2)$

✓ for correct factoring.

d) $x = 3 - \sqrt{2}.$

$x-2 = 1 - \sqrt{2}.$

$(x-2)^2 = (1 - \sqrt{2})^2$

$= 1 - 2\sqrt{2} + 2 = 3 - 2\sqrt{2}.$

✓

✓ for $3 - 2\sqrt{2}.$

e) $2x+1 = 5$

$2x = 4$

$x = 2$

$2x+1 = -5$

$2x = -6$

$x = -3$

✓ for $x = 2$ ✓ for $x = -3$

[For Award 1 for correct working wrong answer].

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Solution of yr 11 2010 Ext1 Half yearly.

Question 2:

a) $\frac{23\pi}{15} = \frac{23 \times 180}{15} = 276$

✓

b) $\operatorname{cosec} 0.8 = \frac{1}{\sin 0.8} = 1.39$ (2 dec. places)

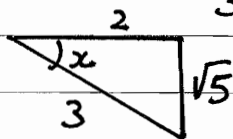
✓

c) $\frac{1 + \cos 120^\circ}{\tan 240} = \frac{1 + (-\frac{1}{2})}{\sqrt{3}} = \frac{1}{2\sqrt{3}}$
 $= \frac{1}{2} \frac{\sqrt{3}}{3} = \frac{\sqrt{3}}{6}$

✓ for correct ratios.

✓ for $\frac{\sqrt{3}}{6}$.

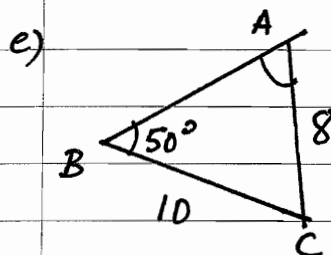
d) $\cos x = -\frac{2}{3}$ $\therefore x$ is in 2nd or 3rd quad.



$\tan x = \pm \frac{\sqrt{5}}{2}$

✓ for correct working ($\sqrt{5}$).

✓ for $\pm \frac{\sqrt{5}}{2}$.



$\frac{\sin A}{10} = \frac{\sin 50}{8}$

$\sin A = \frac{10 \sin 50}{8} = 0.95755$

✓ for correct sine rule

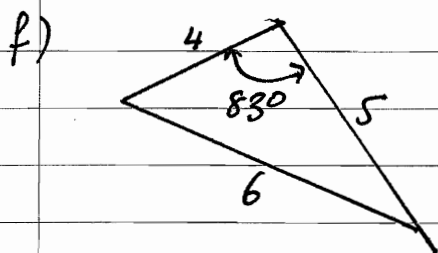
$\therefore \angle A = 73^\circ$, $180 - 73 = 107^\circ$
works since $107 + 50 < 180^\circ$

✓ for 73° .

$\therefore \angle BAC = 73^\circ, 107^\circ$

✓ for obtuse answer.

(Accept: $73^\circ 15'$, $106^\circ 45'$)

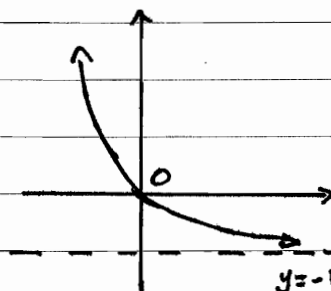
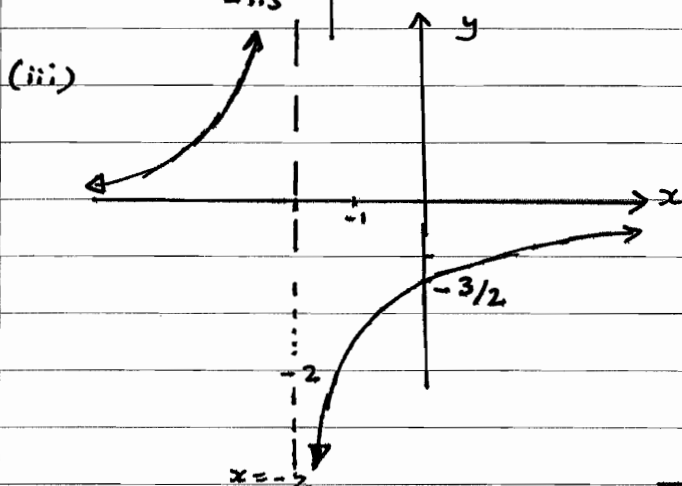
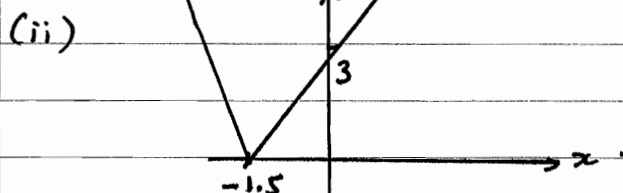
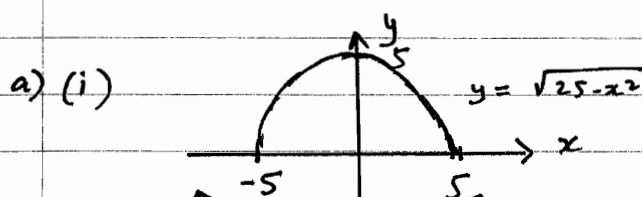


$A = \frac{1}{2} \times 4 \times 5 \sin 83^\circ$
 $= 10 \times \sin 83^\circ = 9.925 \text{ km}^2$
 $= 9.9 \text{ km}^2$

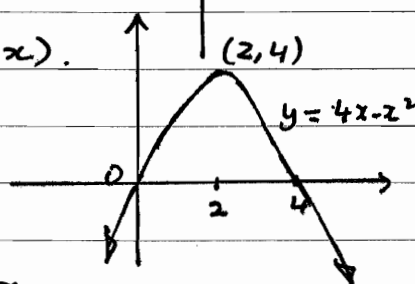
✓ for correct area rule

✓ for correct answer.

Solution of yr 11 Ext 1 Half yearly 2010 -
Question 3.



b) (i) $f(x) = 4x - x^2 = x(4 - x)$.
vertex: $x = 2, y = 4$.



(ii) Domain: All real x .
 $R: y \leq 4$

(iii) $4x - x^2 \geq 0 \therefore 0 \leq x \leq 4$.

(iv) Domain: $0 < x < 4$.

(i)
✓ for shape.
✓ for intercepts.

(ii)
✓ for shape.
✓ for both intercepts

(iii)
✓ for shape.
✓ for asymptote + intercept.

(iv)
✓ for correct shape.
✓ for showing $y = -1$

b) (i)
✓ for shape.
✓ for intercepts and/or vertex.

(ii)
✓

✓

iii)
✓

(iv)
✓ (No equal signs)

Solution of yr 11 2010 Ext 1 Half Yearly.

Question 4:

$$a) (i) \frac{x+3}{2} - \frac{3x-1}{7} = 2. \quad \times 14$$

$$7(x+3) - 2(3x-1) = 28.$$

$$7x + 21 - 6x + 2 = 28.$$

$$x + 23 = 28 \quad \therefore x = 5.$$

(a)

(i)

✓ for correct expansion

✓ for $x = 5$.

$$(ii) 5x^2 - 9x = 2 \quad \therefore 5x^2 - 9x - 2 = 0.$$

$$(5x + 1)(x - 2) = 0.$$

$$x = -\frac{1}{5}, 2.$$

(ii)

✓ for correct method.

✓ for correct answers.

$$(iii) |3x+1| = 4x-2.$$

$$3x+1 = 4x-2.$$

$$3 = x$$

$$3x+1 = -4x+2.$$

$$7x = 1.$$

$$\therefore x = \frac{1}{7}.$$

(iii)

✓ for getting $x=3$.✓ for getting $x = \frac{1}{7}$.
[or ✓ for correct graphical method].

$$\text{test } x=3: 10 = 12-2 \checkmark$$

$$1\frac{3}{7} \neq \frac{4}{7} - 2$$

$\therefore x=3$ is only solution.

✓ for testing to get $x=3$ only.

$$b) (i) \frac{(a-3)(a+3)}{a(a^2-27)} \times \frac{a^2+3a+9}{(a+3)}$$

$$\frac{(a+3)}{a(a+3)(a^2+3a+9)} \times \frac{(a^2+3a+9)}{1}$$

$$= \frac{1}{a}.$$

b) (i)

✓✓ for $\frac{1}{a}$.

(Take away 1 for each error).

$$(ii) \frac{x+2}{x-1} - \frac{2x-1}{x+1} - \frac{6x}{x^2-1} \rightarrow (x+1)(x-1)$$

(ii)

$$= \frac{(x+2)(x+1) - (2x-1)(x-1) - 6x}{(x+1)(x-1)}$$

✓ for correct start.

$$= \frac{x^2 + 3x + 2 - (2x^2 - 3x + 1) - 6x}{x^2 - 1}$$

✓

$$= \frac{-x^2 + 6x + 1 - 6x}{x^2 - 1} = \frac{-1(x^2 - 1)}{x^2 - 1} = -1$$

✓ for -1.

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Solution of yr 11 Ext 1 Half yearly 2010

Question 5

a) (i) $-7 < 5 - 2x < 7$
 $-12 < -2x < 2 \quad \div -2.$
 $6 > x > -1 \quad \therefore -1 < x < 6.$

a) (i)

✓ for correct start.

✓ $-1 < x < 6.$

(ii) $\frac{4x-1}{x-1} - 3 \geq 0.$

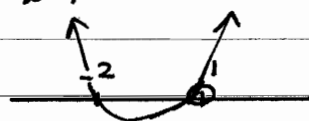
(ii)

$\frac{4x-1-3x+3}{x-1} \geq 0.$

$\frac{x+2}{x-1} \geq 0 \quad x(x-1)^2.$

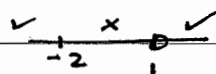
$(x+2)(x-1) \geq 0.$

$x > 1 \text{ or } x \leq -2.$



✓✓ for a correct method.

OR: $4x-1 = 3(x-1)$
 $x = -2.$



$x > 1 \text{ or } x \leq -2.$

✓ for $x > 1 \text{ or } x \leq -$

[Students lose 1 mark for $x \geq 1$]

b) $(2x-1)^4$

b)

$= (2x)^4 + 4(2x)^3(-1) + 6(2x)^2(-1)^2 + 4(2x)(-1)^3 + (-1)^4$

$= 16x^4 - 32x^3 + 24x^2 - 8x + 1.$

✓

✓

c) $\frac{\sqrt{2}+1}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1} = \frac{2+1+2\sqrt{2}}{2-1} = 3+2\sqrt{2}.$

✓ for correct rationalising.

✓ for $3+2\sqrt{2}.$

✓ for $a=3$ and $b=2.$

$a=3. \quad b=2.$

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Solution of Yr 11 Ext 1 Half July 2010

Question 6

a) (i) $\sin 2x = \frac{1}{2}$.

$2x = 30^\circ, 150^\circ, 390^\circ, 510^\circ$.

$x = 15^\circ, 75^\circ, 195^\circ, 255^\circ$.

(ii) $\tan x = \frac{1}{\tan x}$.

$\tan x = 1$

$x = 45^\circ, 180 + 45^\circ$

$x = 45^\circ, 225^\circ$

$\therefore \tan^2 x = 1$.

$\tan x = -1$.

$x = 180 - 45^\circ, 360 - 45^\circ$

$x = 135^\circ, 315^\circ$.

b) LHS $(\tan x + \cot x)^2$

$= \tan^2 x + \cot^2 x + 2$ ✓

$= 1 + \tan^2 x + 1 + \cot^2 x$ ✓

$= \sec^2 x + \operatorname{cosec}^2 x$ ✓

$= \text{RHS}$

or RHS $= \sec^2 x + \operatorname{cosec}^2 x$

$= 1 + \tan^2 x + 1 + \cot^2 x$ ✓

$= \tan^2 x + \cot^2 x + 2$ ✓

$= (\tan x + \cot x)^2$ ✓

$= \text{LHS}$

c) $\cot \theta \times (-\sin \theta)$

$= \frac{\cos \theta}{\sin \theta} \times (-\sin \theta)$

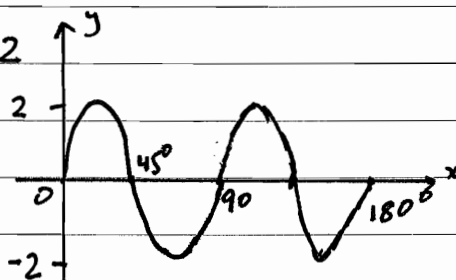
$= -\cos \theta$

d) i) $a = 2$.

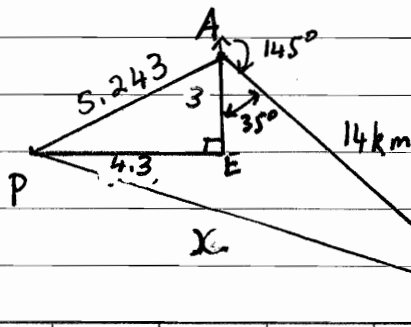
$T = \frac{360}{4} = 90^\circ$

Range = $-2 \leq y \leq 2$

ii)



e)



$AP = \sqrt{4.3^2 + 3^2} = 5.243$

$\tan \angle PAE = \frac{4.3}{3} \Rightarrow \angle PAE = 55^\circ 5' 51''$ ✓

$\therefore \angle PAB = 90^\circ 5' 51''$

$x^2 = 5.243^2 + 14^2 - 2(5.243)(14) \cos(90^\circ 5' 51'')$ ✓ Students have to use cos. rule as $\angle$ is 90.0975°

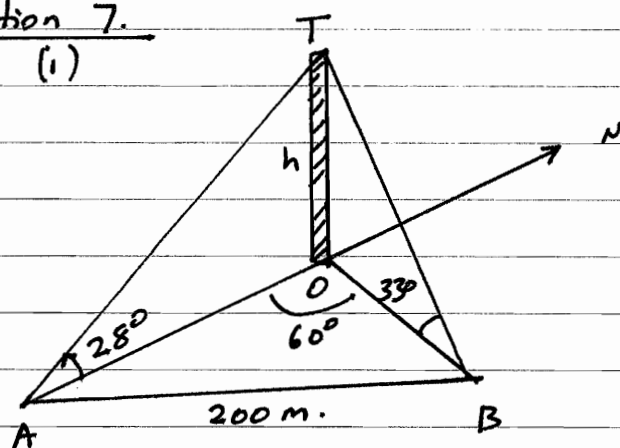
$x = 14.9579$

$14.96 \text{ km (2 dec. pl.)}$ ✓

Solution of yr 11 Ext 1 Half yearly 2010.

Question 7.

(i)



(i)

✓

(ii) $\cot 28^\circ = \frac{OA}{h} \therefore OA = h \cot 28^\circ$

(ii)

✓ for OA

$OB = h \cot 33^\circ$

[smarter: $OA = h \tan 62^\circ$
 $OB = h \tan 57^\circ$]

✓ for OB.

(iii)

(iii)

✓ for cos. rule.

$200^2 = OA^2 + OB^2 - 2OA \cdot OB \cos 60^\circ$

$40000 = h^2 \tan^2 62^\circ + h^2 \tan^2 57^\circ - 2(h \tan 62^\circ)(h \tan 57^\circ) \times \cos 60^\circ$

$$h^2 = \frac{40000}{\tan^2 62^\circ + \tan^2 57^\circ - 2 \tan 62^\circ \tan 57^\circ \times \frac{1}{2}} = 13279.10$$

✓ for h^2

or: $h^2 = \frac{40000}{\cot^2 28^\circ + \cot^2 33^\circ - 2 \cot 28^\circ \cot 33^\circ \times \frac{1}{2}}$

$\therefore h = 115 \text{ m.}$

✓ for 115m.

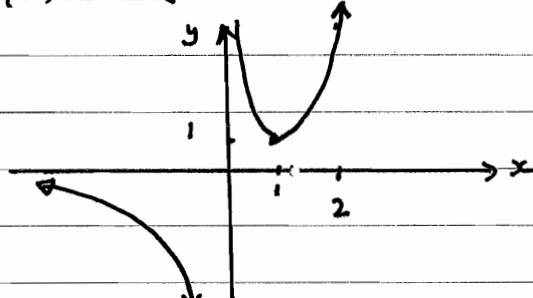
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Solution of Yr 11 Ext 1 Maths Half yearly.

Question 8:

(a) (i) $f(2) = 2(8) - 1 = 15.$

(ii)



(iii) Range: $y < 0$ or $y \geq 1.$

b) $f(n+1) = (n+1)(n+2)(n+3)$

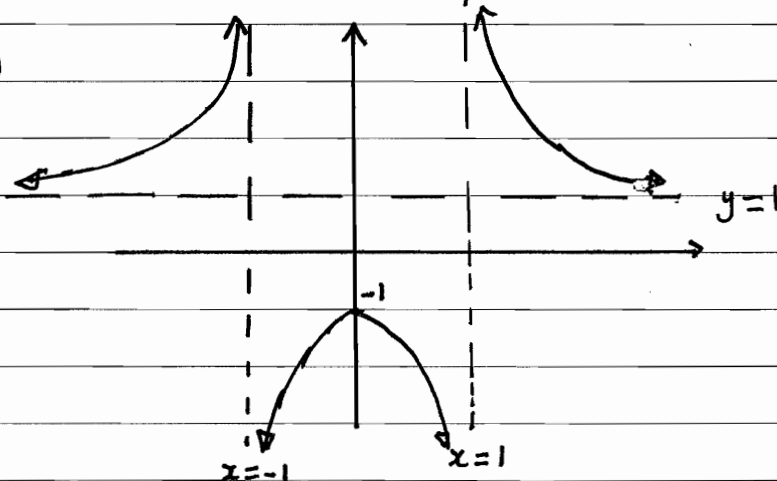
$$\begin{aligned} f(n) + f(n+1) &= n(n+1)(n+2) + (n+1)(n+2)(n+3) \\ &= (n+1)(n+2)(n+n+3) \\ &= (n+1)(n+2)(2n+3) \end{aligned}$$

c) (i) Asymptotes: $y=1.$
 $x=1, x=-1.$

$$\begin{aligned} \text{(ii)} \quad f(-x) &= 1 + \frac{2}{(-x)^2 - 1} \\ &= 1 + \frac{2}{x^2 - 1} = f(x). \end{aligned}$$

$\therefore$ Even function.

(iii)



(iv) $-1 < x < 1.$

d)

$$g(x) = -x^4$$

(i)

✓

(ii)

✓

✓

(iii)

✓

b)

✓ for correct $f(n+1)$

✓ for final factored answer.

c) (i)

✓ for $y=1$

✓ for both $x=\pm 1$

(ii)

✓ ✓

✓ ✓ ✓

Asymptotes have to be shown clearly esp. $y=1.$ The 3 parts of the graph have to be clearly drawn to get the 3 marks.

✓ only for $-1 < x < 1$ (question states Hence, or otherwise)

✓